1B Frequent Words Problem

Solve the Frequent Words Problem.

Input: A string Text and an integer k.

Output: All most frequent k-mers in Text.

Test Cases

Case 1
Description: The sample dataset. TGG occurs four times and no other 3-mer occurs more than twice, so the answer is a single k-mer with a comfortable margin. Case 5 is the one that checks the boundary at the start of Text.

Input:

```
TGGACGTTGGCCCAGCTGGTCCCACGTGGT
3
```

Output:

```
TGG
```

Case 2
Description: This dataset checks whether you count the last k-mer in Text (TTTT in this example). If you do not count the last k-mer, you will report additional "most frequent" 4-mers alongside TTTT.

Input:

```
CGTTTTGAACATTTTCAACAAGTTTTGCAACATTTT
4
```

Output:

```
TTTT
```

Case 3
Description: This dataset checks that your code handles overlapping occurrences of a pattern. For example, AACAACAA contains two overlapping occurrences of AACAA; if your code counts them as a single occurrence, it will fail this case.

Input:

```
GTTGGGTTGGAACAACAACAACAACAAGTTGGGTTGG
5
```

Output:

```
AACAA
```

Case 4

Description: This dataset checks that you handle ties correctly: your code should output ALL most frequent k-mers, not just one. Here both CATG and GCAT occur three times, so both should be output.

Input:

```
ACGTTGCATGTCGCATGATGCATGAGAGCT
4
```

Output:

```
CATG GCAT
```

Case 5

Description: This dataset checks whether you count the k-mer at the very START of Text. TTC occurs twice, at positions 0 and 12, and every other 3-mer occurs once, so TTC is the unique answer. If your loop begins at position 1 rather than 0, TTC drops to one occurrence, every 3-mer ties, and you will report all twelve of them instead of TTC alone.

Input:

```
TTCCTCATGCAATTC
3
```

Output:

```
TTC
```

Case 6

Description: This is the full-size dataset we actually grade your code on. There are no special edge cases hiding here, so if your code handled the smaller tests above, it should handle this one too. Nice work making it this far!

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